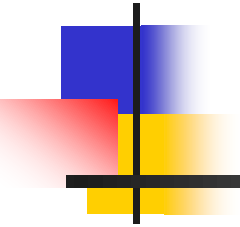


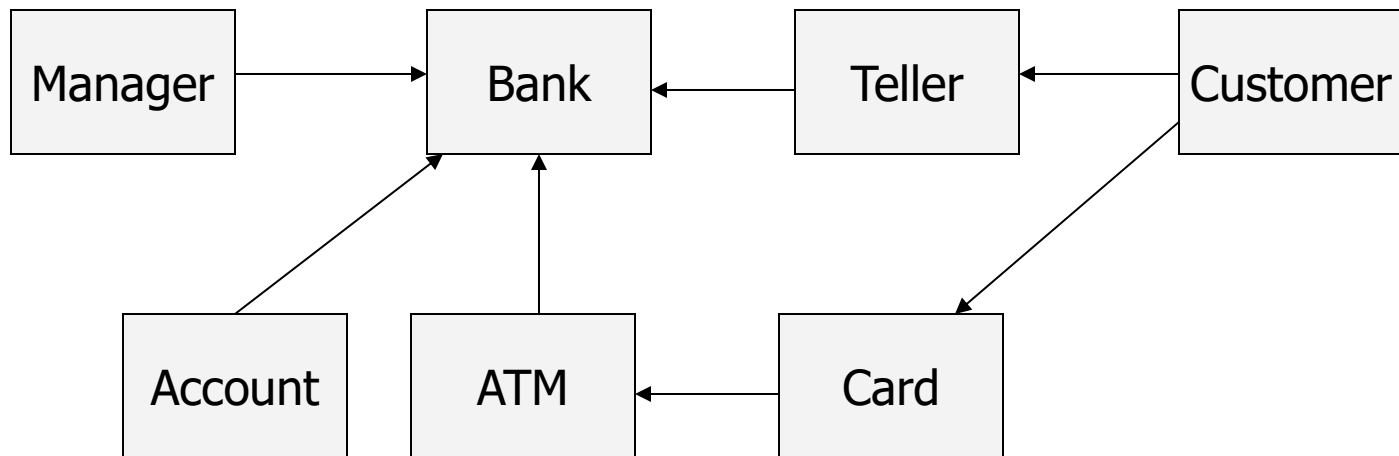
1. Introduction to Object Orientation

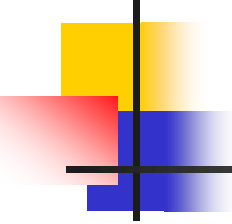


What is Object Orientation?

- Organizing software as a collection of discrete objects that incorporate both data structure (attributes) and behavior.

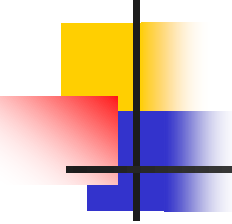
Ex: Banking Software





Objects and Classes

- Object is an instance of a Class. That is, every Object has a Class.
- Object has a unique identity. Two objects of a same class are distinguishable.
- A Class describes a group of Objects with similar properties (attributes), common behavior (operation).

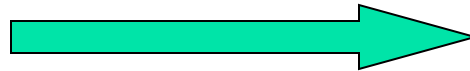


Objects and Classes

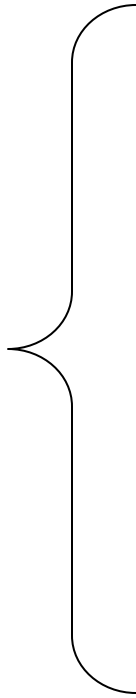
Class



Instantiation



Objects

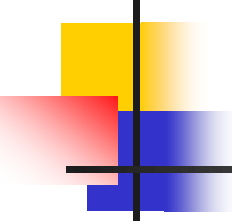


Bus

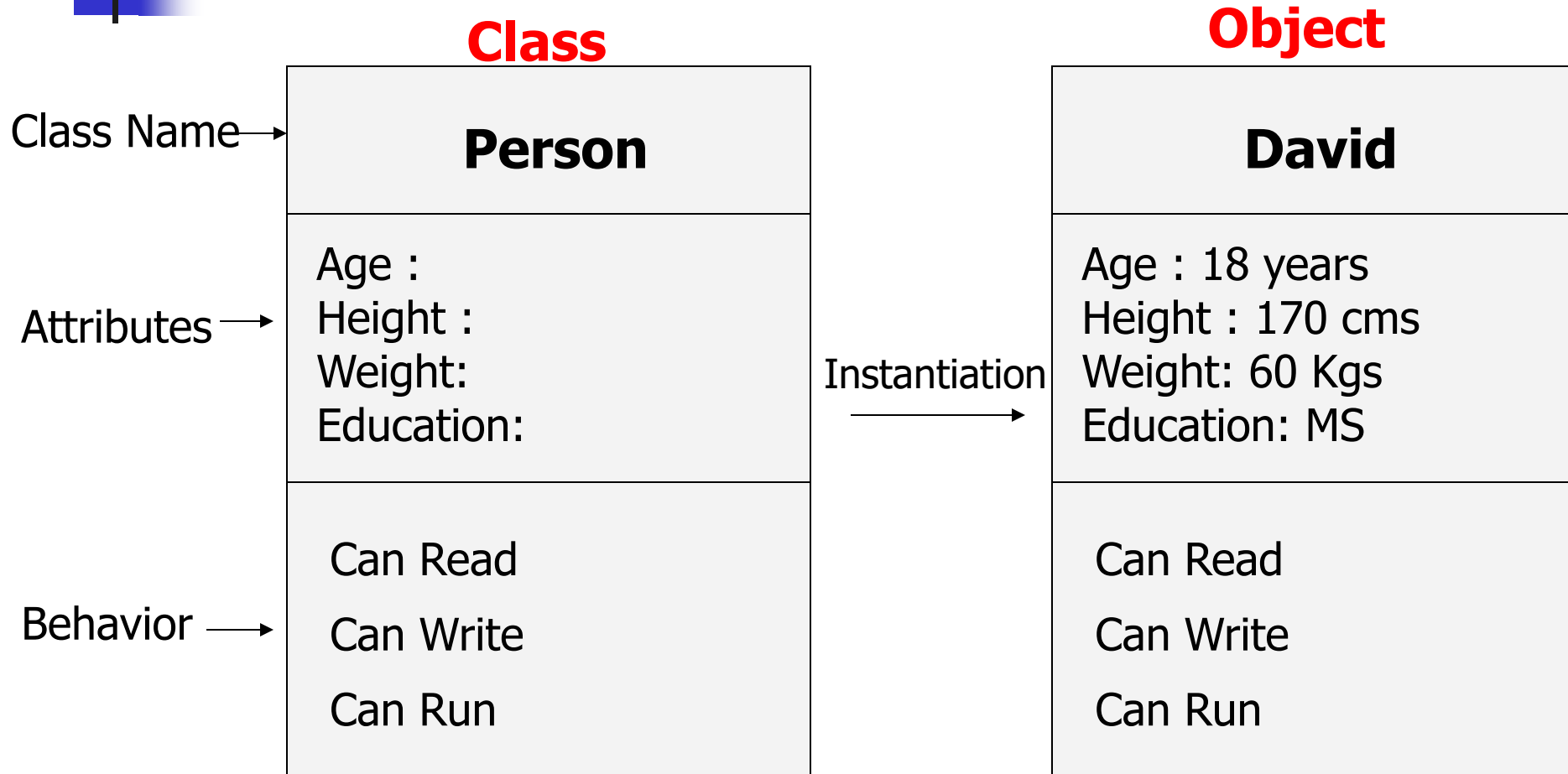
Car

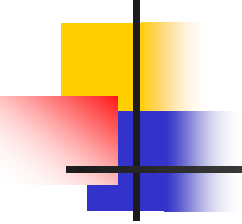
Truck

Motor
Cycle



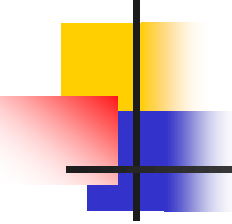
A Class of an Object





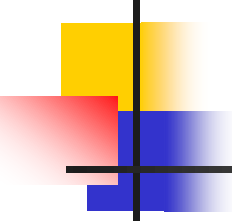
Three OOP Principles

1. Encapsulation
2. Inheritance
3. Polymorphism

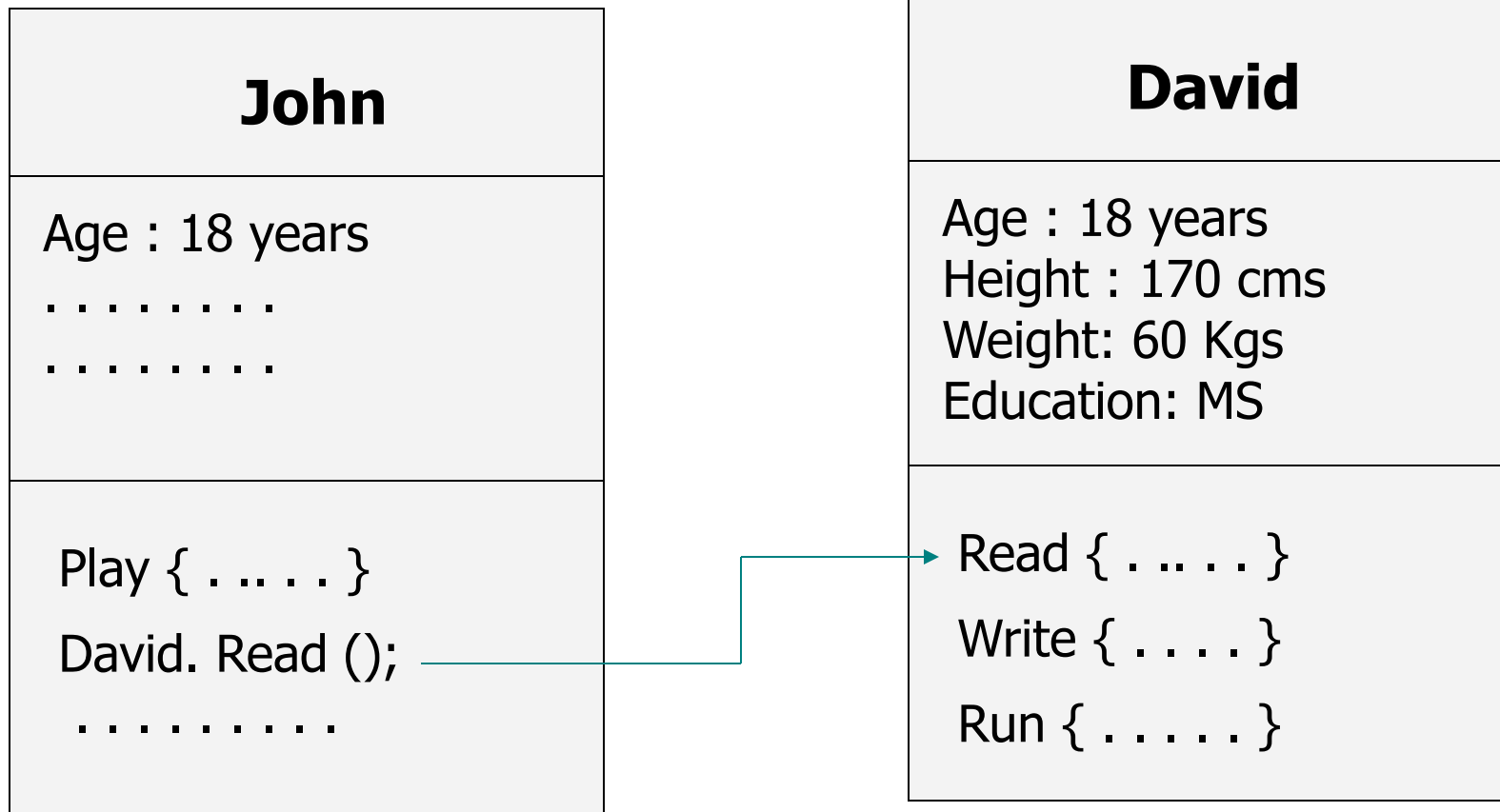


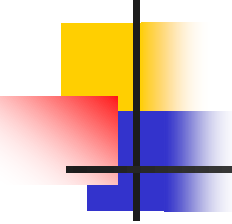
Encapsulation

- Also known as ***Data Hiding***.
- Separating the external aspects of an object, which are accessible to other objects, from the internal implementation details of the object, which are hidden from other objects.
- Since data and behavior are combined in a single entity, this makes encapsulation cleaner and more powerful.



Encapsulation

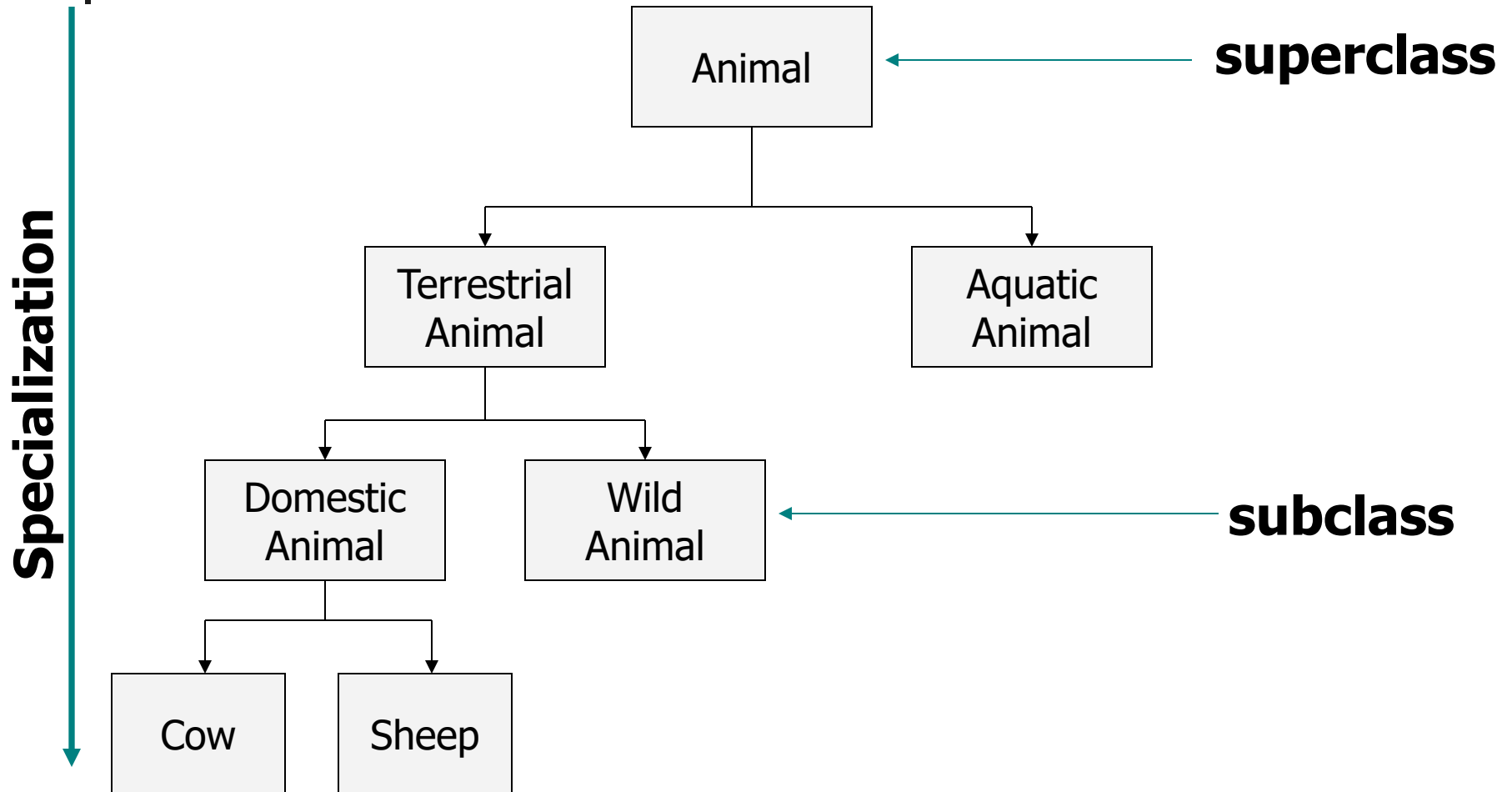


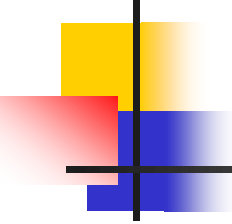


Inheritance

- Inheritance is a powerful abstraction for sharing similarities among classes while preserving their differences.
- This is the relationship between a class and one or more refined versions of it.
 - The class being refined is called the *superclass*
 - Each refined version is called *subclass*

Inheritance

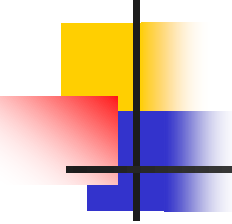




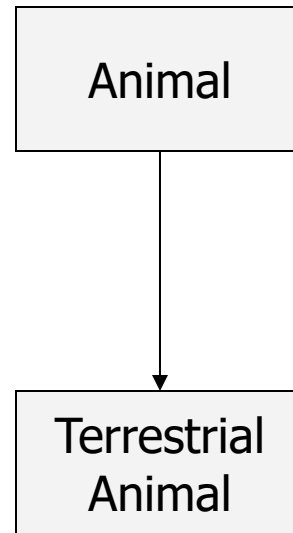
Inheritance

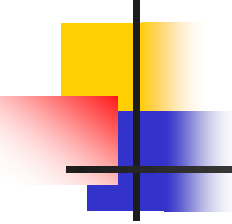
Types of Inheritance:

- Single Inheritance
- Multilevel Inheritance
- Multiple Inheritance

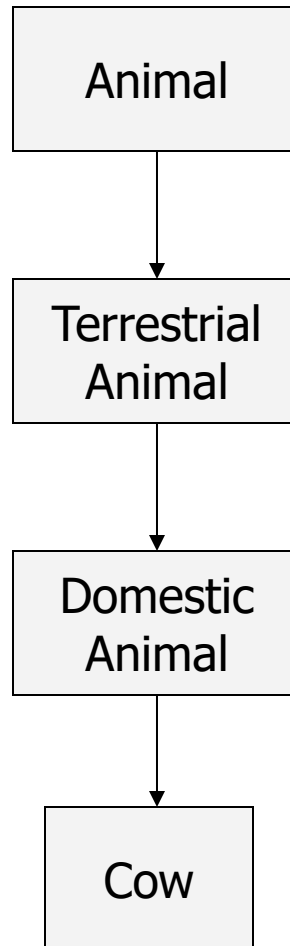


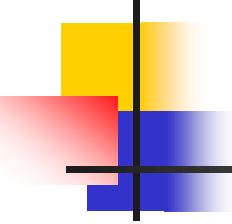
Single Inheritance



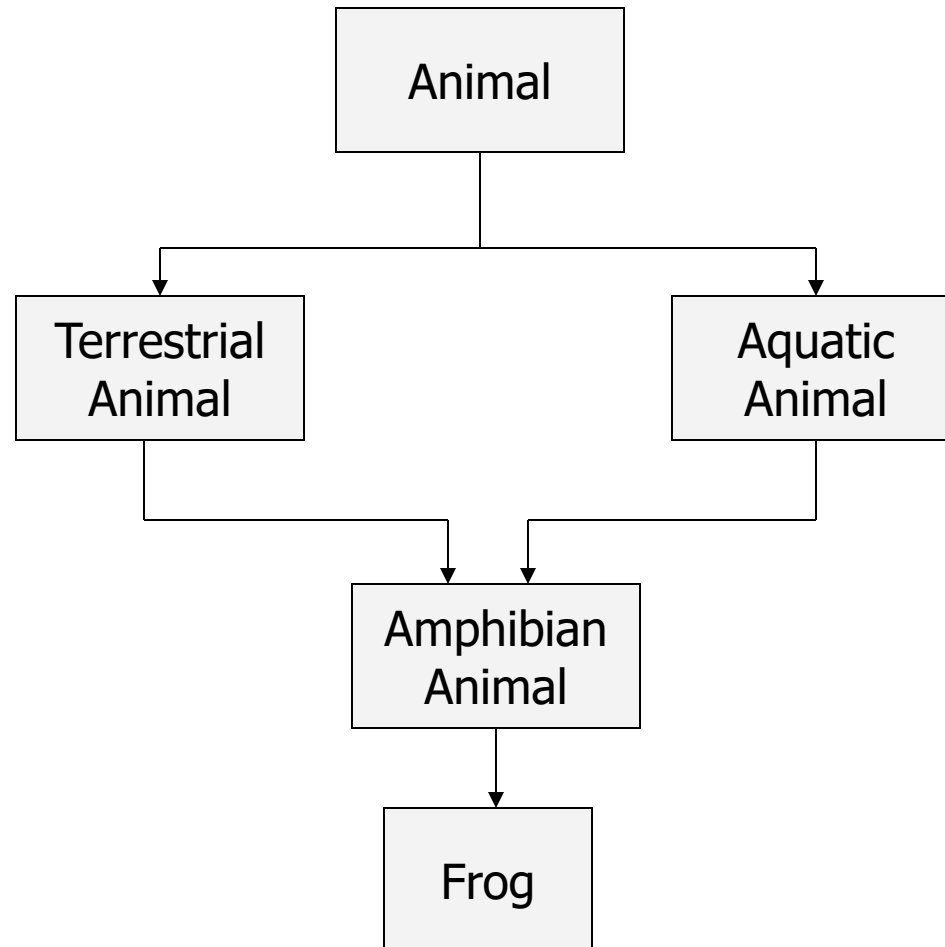


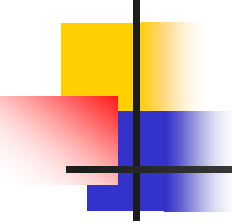
Multilevel Inheritance





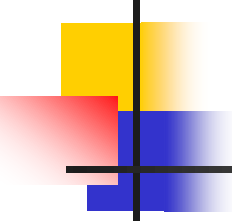
Multiple Inheritance





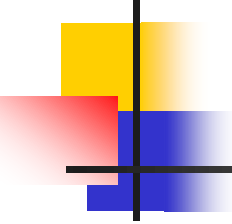
Polymorphism

- *Polymorphism* means the ability to take more than one form.
- An operation may exhibit different behaviors in different instances. The behavior depends on the data types used in the operation.
- Polymorphism is extensively used in implementing Inheritance.



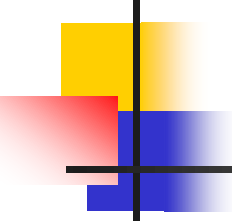
Object-Oriented Programming

- OOP is a programming methodology that helps organize complex programs through the use of inheritance, encapsulation and polymorphism.



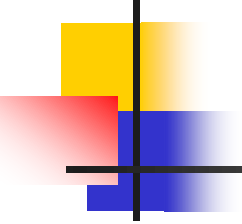
Features of OOP

- Emphasis on data rather than procedure.
- Programs are divided into Objects.
- Data is hidden and cannot be accessed by external functions.
- Objects can communicate with each other through public interfaces.
- New data and methods can be easily added whenever necessary.
- Follows bottom-up approach.



OOP versus Procedural approach

- Easy to understand, maintain, modify.
- 100% bug free software possible
- Helps code reuse
- Bottom-up approach
- Difficult
- Impossible to make 100% bug free.
- Code reuse difficult.
- Top-down approach.



References

- Herbert Schildt, Java 2: The Complete Reference, 7th Edition., McGraw-Hill Publishing Company Inc
- "The Java Tutorial", found online at <https://docs.oracle.com/javase/tutorial/tutorialLearningPaths.html>